

Class Activity – 1

Scenario 1: Student Academic Performance Dashboard

R program:

```
library(ggplot2)

student <- data.frame(

  StudentID = 1:10,

  Attendance = c(85,90,78,95,88,70,80,92,75,89),

  Marks = c(80,92,70,95,85,60,75,94,68,88),

  Department = c("CSE","AI","IT","ECE","CSE","AI","IT","ECE","CSE","AI")

)

ggplot(student, aes(x=Attendance, y=Marks, color=Department)) +

  geom_point(size=4) +

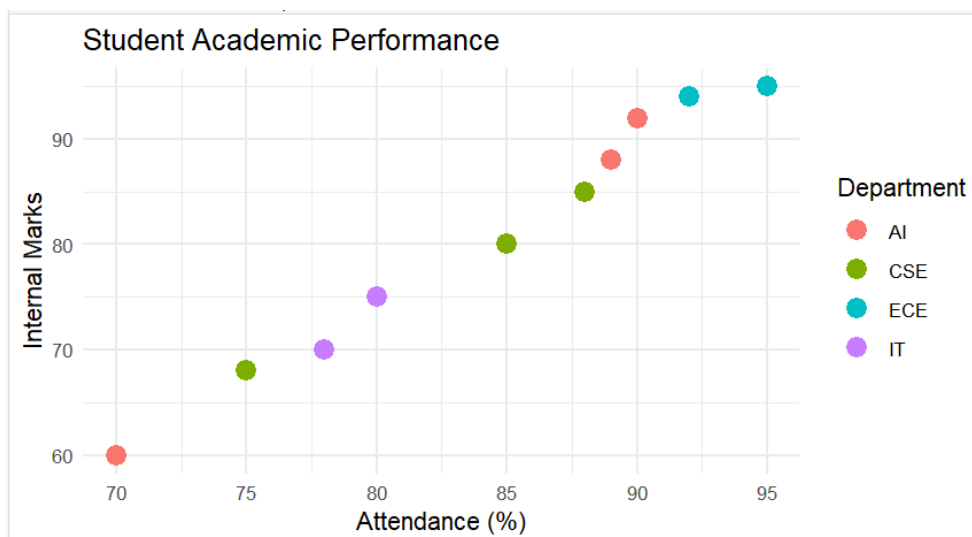
  labs(title="Student Academic Performance",

        x="Attendance (%)",

        y="Internal Marks",

        color="Department") +

  theme_minimal()
```



Python Program:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

student = pd.DataFrame({

    "StudentID":[1,2,3,4,5,6,7,8,9,10],

    "Attendance":[85,90,78,95,88,70,80,92,75,89],

    "Marks":[80,92,70,95,85,60,75,94,68,88],

    "Department":["CSE", "AI", "IT", "ECE", "CSE", "AI", "IT", "ECE", "CSE", "AI"]

})

plt.figure(figsize=(8,6))

sns.scatterplot(data=student,

                x="Attendance",

                y="Marks",

                hue="Department",

                s=100)

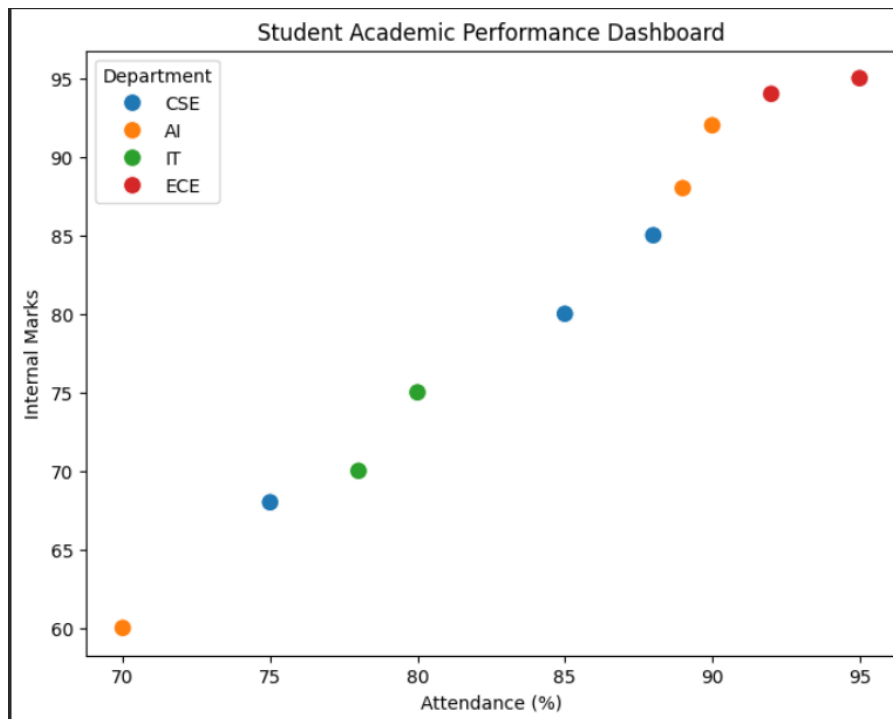
plt.title("Student Academic Performance Dashboard")

plt.xlabel("Attendance (%)")

plt.ylabel("Internal Marks")

plt.legend(title="Department")

plt.show()
```



Scenario 2: Electric Vehicle Market Analysis

R program:

```
library(ggplot2)

ev <- data.frame(

  Model=c("A","B","C","D","E","F"),

  Battery=c(40,50,60,75,90,100),

  Price=c(12,15,20,28,35,40),

  Range=c(250,320,380,450,520,600),

  Manufacturer=c("Tesla","BYD","Tata","Tesla","BYD","Tata")

)

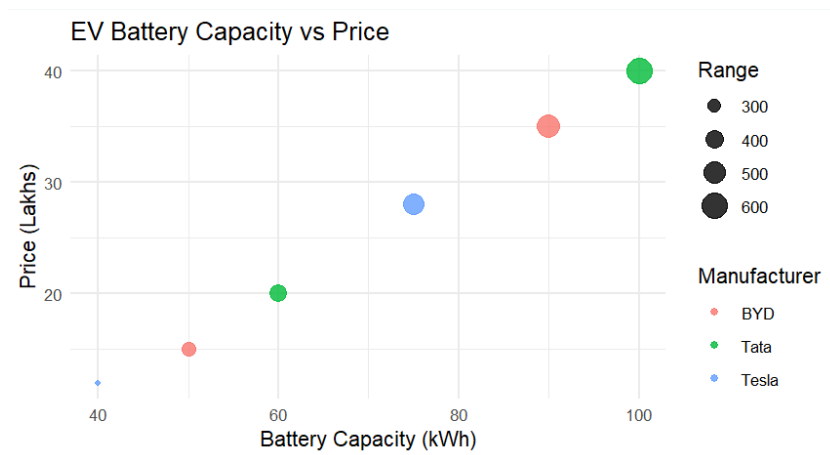
ggplot(ev, aes(x=Battery,

  y=Price,

  size=Range,

  color=Manufacturer)) +
```

```
geom_point(alpha=0.8) +
labs(title="EV Battery Capacity vs Price",
      x="Battery Capacity (kWh)",
      y="Price (Lakhs)") +
theme_minimal()
```



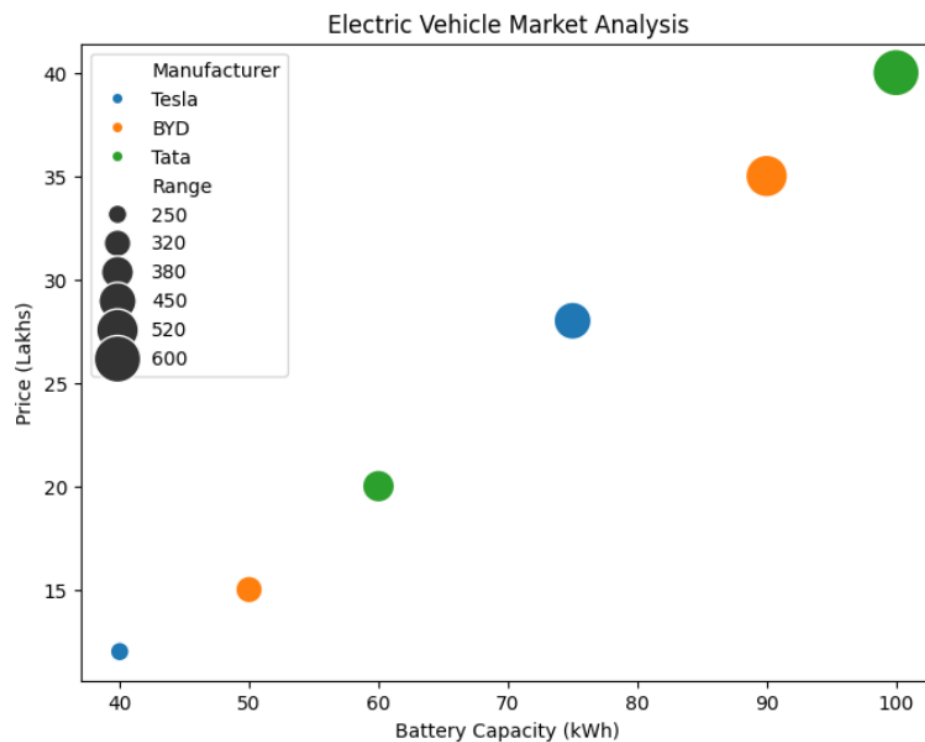
Python Program:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

ev = pd.DataFrame({
    "Model":["A","B","C","D","E","F"],
    "Battery":[40,50,60,75,90,100],
    "Price":[12,15,20,28,35,40],
    "Range":[250,320,380,450,520,600],
    "Manufacturer":["Tesla","BYD","Tata","Tesla","BYD","Tata"]
})

plt.figure(figsize=(8,6))
```

```
sns.scatterplot(data=ev,  
                x="Battery",  
                y="Price",  
                hue="Manufacturer",  
                size="Range",  
                sizes=(100,600))  
  
plt.title("Electric Vehicle Market Analysis")  
plt.xlabel("Battery Capacity (kWh)")  
plt.ylabel("Price (Lakhs)")  
plt.show()
```



Scenario 3: Hospital Patient Admission Analysis

R program:

```
library(ggplot2)

hospital <- data.frame(

  Department=c("Cardiology","Neurology","Orthopedics","Pediatrics"),

  Patients=c(150,120,180,160),

  Cost=c(50000,65000,45000,30000)

)

ggplot(hospital,

  aes(x=Department,

      y=Patients,

      fill=Department)) +

  geom_bar(stat="identity") +

  geom_text(aes(label=Cost),

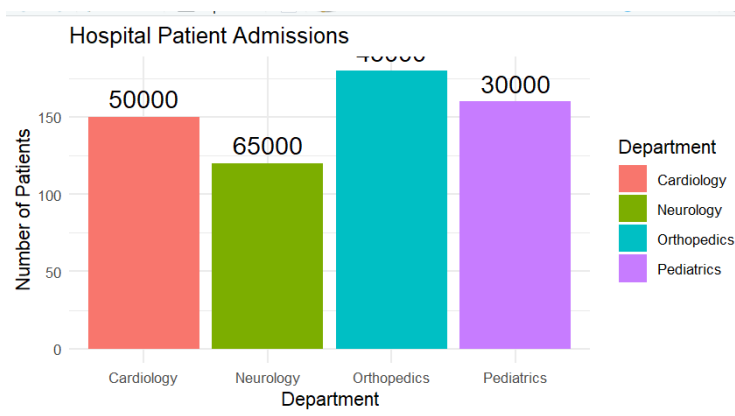
    vjust=-0.5,

    size=5) +

  labs(title="Hospital Patient Admissions",

    y="Number of Patients") +

  theme_minimal()
```



Python Program:

```
import pandas as pd

import matplotlib.pyplot as plt

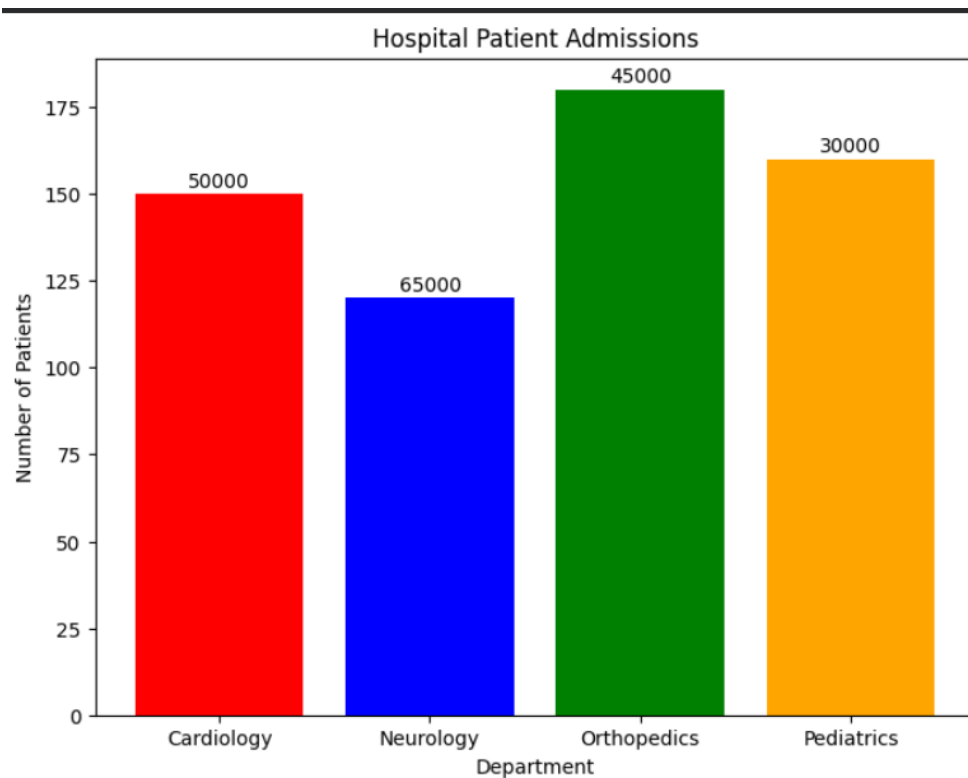
hospital = pd.DataFrame({
    "Department":["Cardiology","Neurology","Orthopedics","Pediatrics"],
    "Patients":[150,120,180,160],
    "Cost":[50000,65000,45000,30000]
})

plt.figure(figsize=(8,6))

bars = plt.bar(hospital["Department"],
               hospital["Patients"],
               color=["red","blue","green","orange"])

for i, bar in enumerate(bars):
    plt.text(bar.get_x()+bar.get_width()/2,
             bar.get_height()+2,
             hospital["Cost"][i],
             ha='center')

plt.title("Hospital Patient Admissions")
plt.xlabel("Department")
plt.ylabel("Number of Patients")
plt.show()
```



Scenario 4: Weather Monitoring System

R program:

```
library(ggplot2)

weather <- data.frame(

  Date=rep(1:10,3),

  Temperature=c(30,31,32,33,32,31,30,29,30,31,

                28,29,30,31,32,33,34,33,32,31,

                25,26,27,28,29,30,29,28,27,26),

  City=rep(c("Chennai","Delhi","Bangalore"),each=10)

)

ggplot(weather,

  aes(x=Date,
```

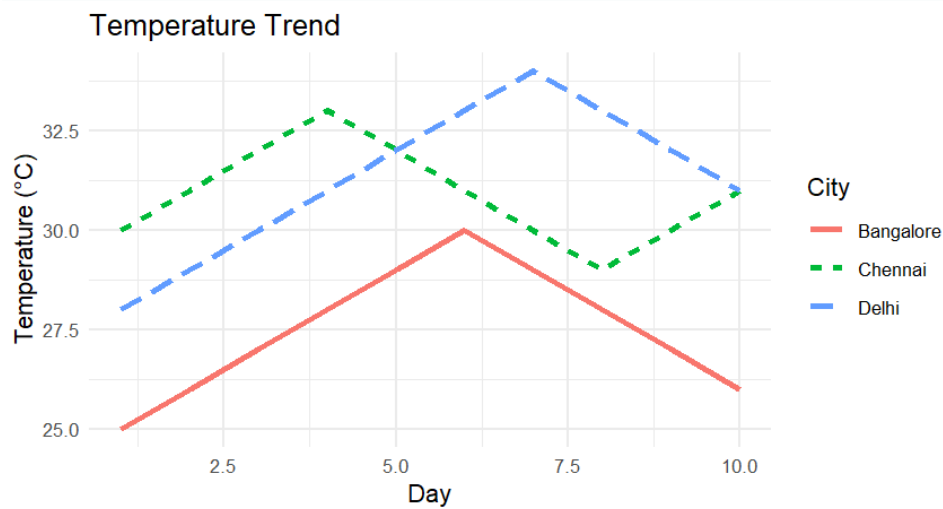
```

y=Temperature,

color=City,

linetype=City)) +
geom_line(size=1.2) +
labs(title="Temperature Trend",
x="Day",
y="Temperature (°C)") +
theme_minimal()

```



Python Program:

```

import pandas as pd
import matplotlib.pyplot as plt

weather = pd.DataFrame({
    "Date":list(range(1,11))*3,
    "Temperature":[30,31,32,33,32,31,30,29,30,31,
                  28,29,30,31,32,33,34,33,32,31,
                  25,26,27,28,29,30,29,28,27,26],
    "City":["Chennai"]*10 + ["Delhi"]*10 + ["Bangalore"]*10
})

```

```
})
```

```
plt.figure(figsize=(8,6))
```

```
styles = {
```

```
    "Chennai": "-",
```

```
    "Delhi": "--",
```

```
    "Bangalore": ":"
```

```
}
```

```
for city in weather["City"].unique():
```

```
    data = weather[weather["City"]==city]
```

```
    plt.plot(data["Date"],
```

```
            data["Temperature"],
```

```
            linestyle=styles[city],
```

```
            linewidth=2,
```

```
            label=city)
```

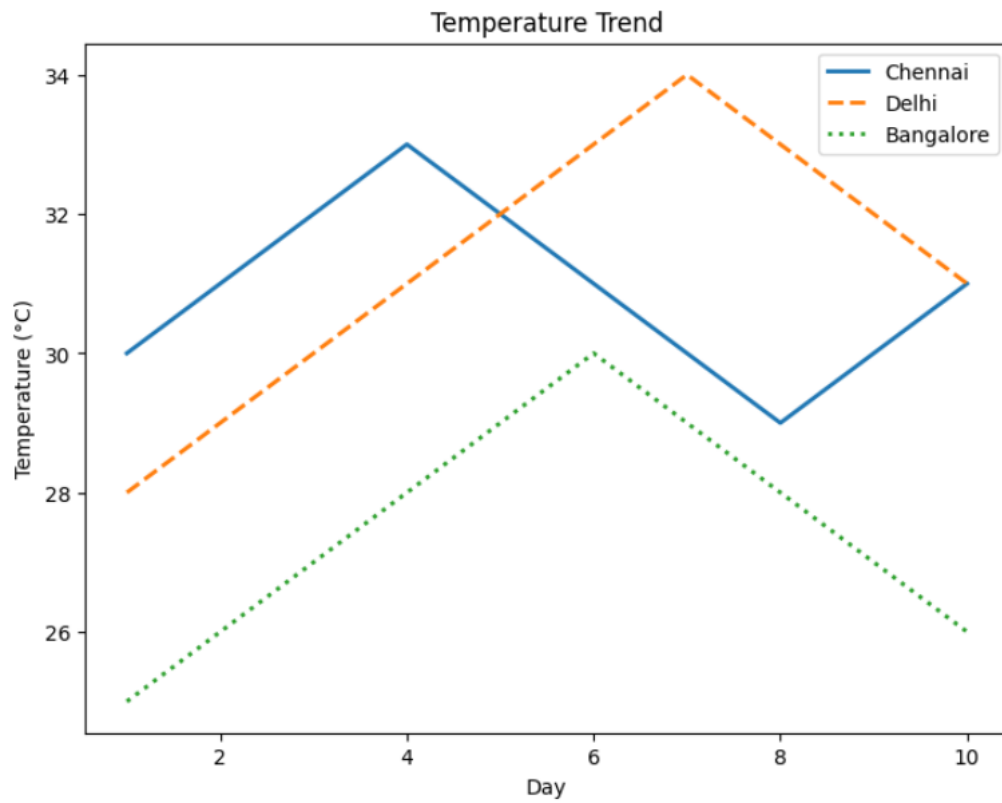
```
plt.title("Temperature Trend")
```

```
plt.xlabel("Day")
```

```
plt.ylabel("Temperature (°C)")
```

```
plt.legend()
```

```
plt.show()
```



Scenario 5: Supermarket Sales Analysis

R program:

```
library(ggplot2)

sales <- data.frame(

  Month=rep("January",9),

  Category=rep(c("Food","Drinks","Snacks"),3),

  Sales=c(120,90,80,

          140,100,95,

          130,85,100),

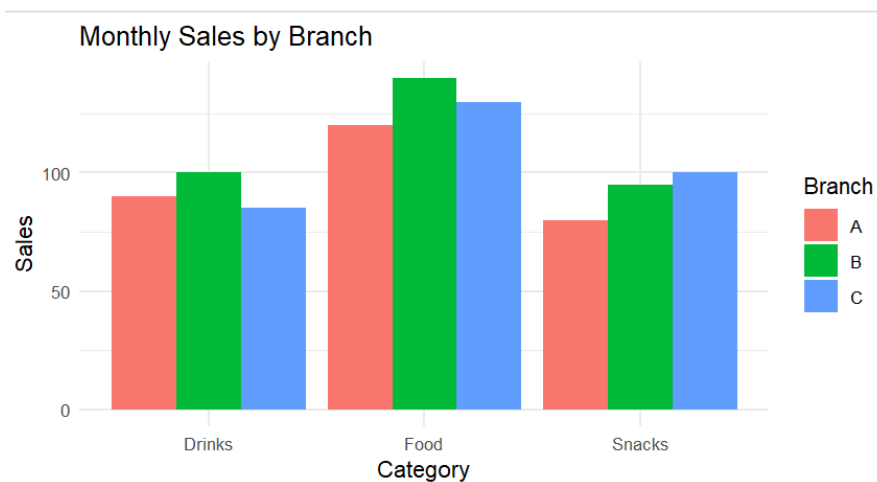
  Branch=rep(c("A","B","C"),each=3)

)
```

```

ggplot(sales,
      aes(x=Category,
          y=Sales,
          fill=Branch)) +
  geom_bar(stat="identity",
          position="dodge") +
  labs(title="Monthly Sales by Branch") +
  theme_minimal()

```



Python Program:

```

import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

sales = pd.DataFrame({
    "Category":["Food","Drinks","Snacks"]*3,
    "Sales":[120,90,80,
             140,100,95,
             130,85,100],
    "Branch":["A"]*3 + ["B"]*3 + ["C"]*3

```

```
})
```

```
categories = ["Food", "Drinks", "Snacks"]
```

```
branches = ["A", "B", "C"]
```

```
x = np.arange(len(categories))
```

```
width = 0.25
```

```
plt.figure(figsize=(8,6))
```

```
for i, branch in enumerate(branches):
```

```
    values = sales[sales["Branch"]==branch]["Sales"]
```

```
    plt.bar(x+i*width,
```

```
           values,
```

```
           width,
```

```
           label=branch)
```

```
plt.xticks(x+width, categories)
```

```
plt.title("Monthly Sales by Branch")
```

```
plt.xlabel("Product Category")
```

```
plt.ylabel("Sales Amount")
```

```
plt.legend(title="Branch")
```

```
plt.show()
```



Scenario 6: Employee Salary Distribution

R program:

```
library(ggplot2)

set.seed(100)

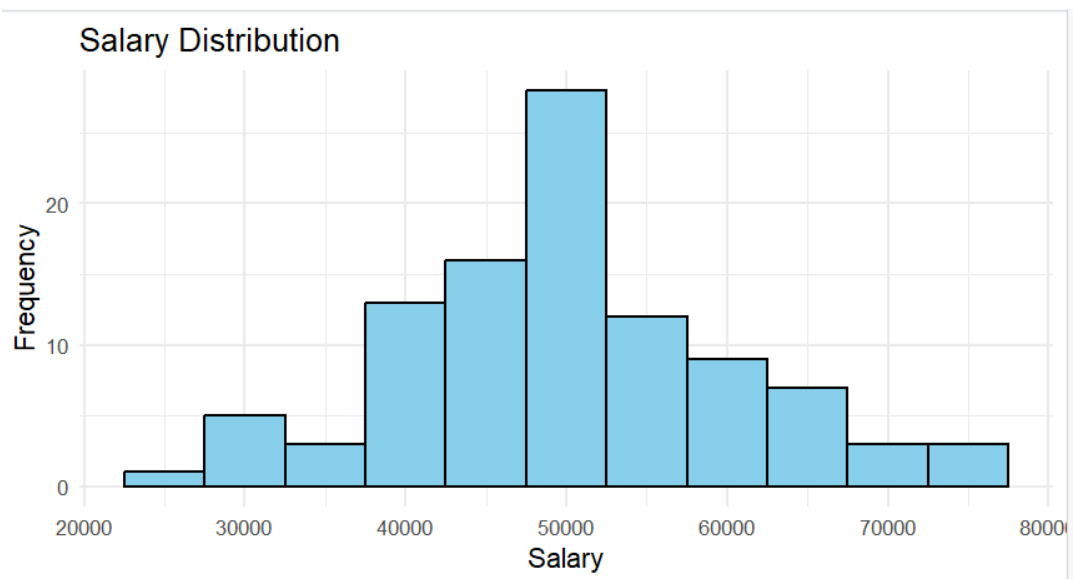
employee <- data.frame(
  EmployeeID=1:100,
  Salary=rnorm(100,50000,10000),
  Experience=sample(1:20,100,replace=TRUE),
  Department=sample(c("IT","HR","Sales"),100,replace=TRUE)
)

ggplot(employee,
  aes(x=Salary)) +
```

```

geom_histogram(binwidth=5000,
               fill="skyblue",
               color="black") +
labs(title="Salary Distribution",
     x="Salary",
     y="Frequency") +
theme_minimal()

```



Python Program:

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(100)

employee = pd.DataFrame({
    "EmployeeID":range(1,101),
    "Salary":np.random.normal(50000,10000,100),

```

```
"Experience":np.random.randint(1,21,100),  
"Department":np.random.choice(["IT","HR","Sales"],100)  
})
```

```
plt.figure(figsize=(8,6))
```

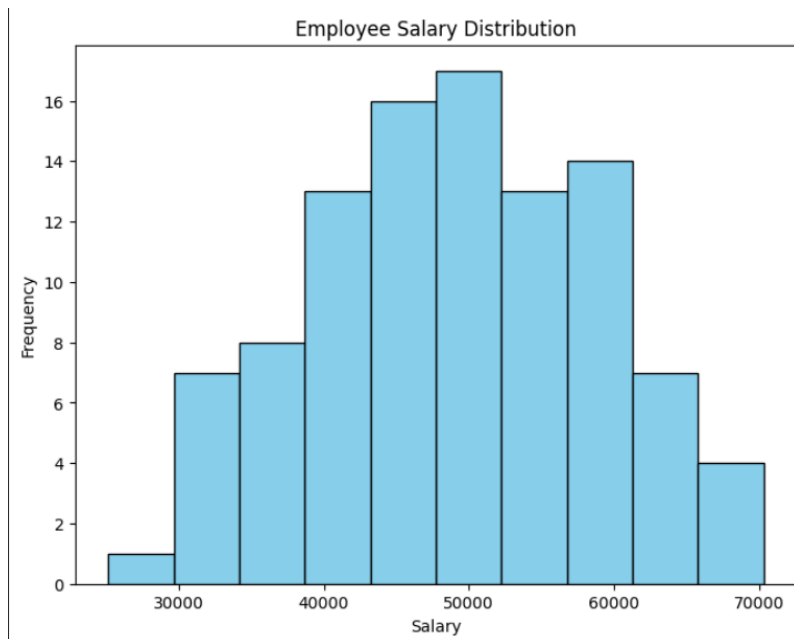
```
plt.hist(employee["Salary"],  
        bins=10,  
        color="skyblue",  
        edgecolor="black")
```

```
plt.title("Employee Salary Distribution")
```

```
plt.xlabel("Salary")
```

```
plt.ylabel("Frequency")
```

```
plt.show()
```



Scenario 7: Smartphone Market Comparison

R program:

```
library(ggplot2)

# Dataset

smartphone <- data.frame(

  Brand = c("Samsung", "Apple", "OnePlus", "Xiaomi", "Realme", "Samsung", "Apple", "OnePlus"),

  RAM = c(4, 6, 8, 6, 8, 12, 8, 12),

  ProcessorSpeed = c(2.0, 2.5, 3.0, 2.2, 2.8, 3.2, 3.1, 3.3),

  Price = c(15000, 60000, 35000, 20000, 25000, 70000, 65000, 55000),

  Storage = c(64, 128, 128, 64, 256, 256, 512, 256)

)

# Plot

ggplot(smartphone,

  aes(x = RAM,

    y = Price,

    shape = Brand,

    size = Storage,

    color = ProcessorSpeed)) +

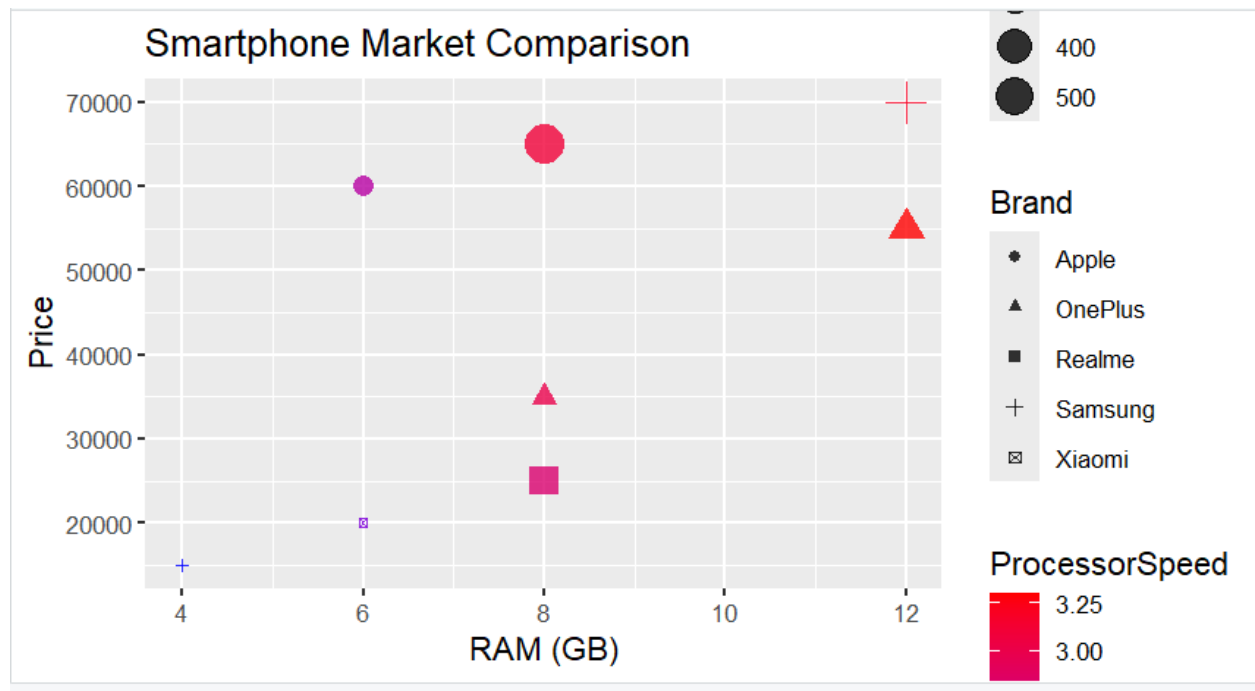
  geom_point(alpha = 0.8) +

  scale_color_gradient(low = "blue", high = "red") +

  labs(title = "Smartphone Market Comparison",

    x = "RAM (GB)",

    y = "Price")
```



Scenario 8: Online Movie Rating Analysis

R program:

```
library(ggplot2)
```

```
# Dataset
```

```
movies <- data.frame(
  Genre = c("Action", "Comedy", "Drama", "Horror", "Sci-Fi"),
  AverageRating = c(8.1, 7.4, 8.8, 6.9, 8.5),
  NumberOfReviews = c(5000, 4200, 6000, 3500, 4800),
  ReleaseYear = c(2025, 2025, 2025, 2025, 2025)
)
```

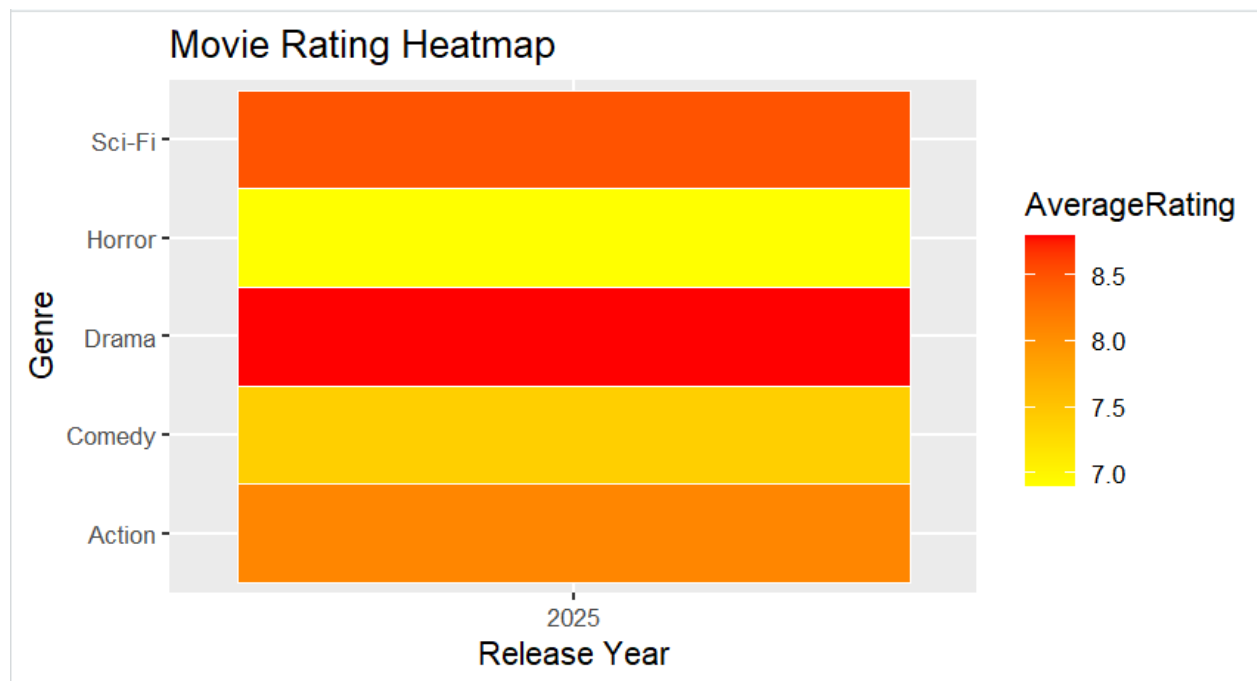
```
# Heatmap
```

```
ggplot(movies,
```

```

aes(x = factor(ReleaseYear),
    y = Genre,
    fill = AverageRating)) +
geom_tile(color = "white") +
scale_fill_gradient(low = "yellow", high = "red") +
labs(title = "Movie Rating Heatmap",
     x = "Release Year",
     y = "Genre")

```



Scenario 9: COVID-19 State-wise Analysis

R program:

```
library(ggplot2)

# Dataset

covid <- data.frame(

  State = c("Tamil Nadu", "Kerala", "Karnataka", "Maharashtra", "Delhi"),

  ActiveCases = c(1200, 1800, 900, 2500, 700),

  VaccinationPercentage = c(92, 90, 91, 89, 95),

  Population = c(78, 35, 68, 125, 20)

)

# Tile Plot

ggplot(covid,

  aes(x = 1,

    y = reorder(State, ActiveCases),

    fill = ActiveCases)) +

  geom_tile(color = "white") +

  scale_fill_gradient(low = "lightblue", high = "darkred") +

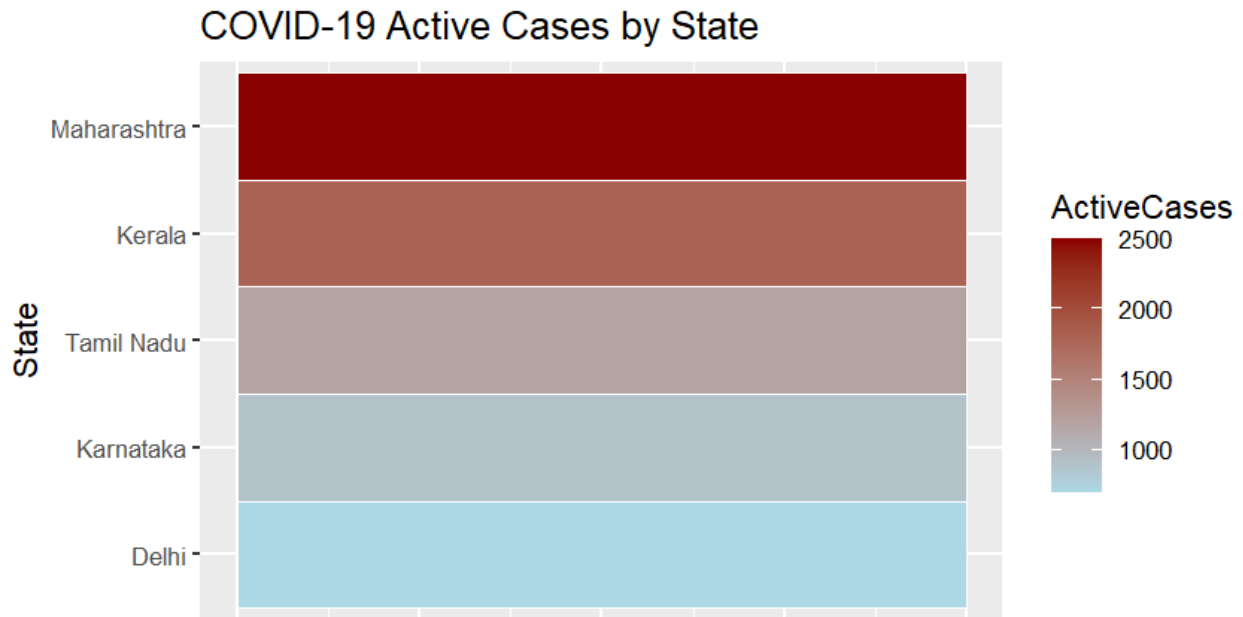
  labs(title = "COVID-19 Active Cases by State",

    x = "",

    y = "State") +

  theme(axis.text.x = element_blank(),

    axis.ticks.x = element_blank())
```



Scenario 10: Iris Flower Classification

R program:

```
library(ggplot2)
```

```
# Built-in Dataset
```

```
data(iris)
```

```
# Scatter Plot
```

```
ggplot(iris,  
  aes(x = Sepal.Length,  
    y = Petal.Length,  
    color = Species,  
    shape = Species,  
    size = Sepal.Width)) +
```

```
geom_point(alpha = 0.8) +  
labs(title = "Iris Flower Classification",  
      x = "Sepal Length",  
      y = "Petal Length")
```

